

Mine Gases —

Q. Calculate the % atmospheric air & various damps present in a mine which gave the following analysis with the air analysis results.

$$O_2 \longrightarrow 16.52\%$$

$$N_2 \longrightarrow 77.93\%$$

$$CO_2 \longrightarrow 3.1\%$$

$$CH_4 \longrightarrow 2.44\%$$

$$CO \longrightarrow 0.01\%$$

$$O_2 (20.93\%) \longrightarrow N_2 (79.04\%)$$

$$O_2 (16.52\%) \longrightarrow \left(\frac{79.04}{20.93} \times 16.52 \right) \%$$

$$N_2 (62.39\%)$$

$$\left\{ \begin{array}{l} \text{Atmospheric air composition.} \\ N_2 \longrightarrow 79.04\% \\ O_2 \longrightarrow 20.93\% \\ CO_2 \longrightarrow 0.03\% \end{array} \right.$$

Similarly, CO_2 —

$$CO_2 = \frac{0.03}{20.93} \times 16.52 = 0.024\%$$

$$\begin{aligned} \% \text{ of atmospheric air} &= (16.52 + 62.39 + 0.024)\% \\ &= 78.93\% \end{aligned}$$

$$\text{Excessive } CO_2 = 3.1 - 0.024 = 3.076\%$$

$$\text{Excessive } N_2 = 77.93 - 62.39 = 15.54\%$$

$$\begin{aligned} \text{Black damp} &= N_2 + CO_2 \\ &= 3.076 + 15.54 \\ &= 18.616\% \end{aligned}$$

Heat & Humidity —

$$pV = nRT$$

$$p \propto T$$

⇒ more compression → more rise in temperature
 (potential energy) (thermal energy)

* If no exchange (loss or gain) in shaft of heat or moisture, the compression occurs adiabatically —

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

T = Kelvin

$$\gamma = \frac{C_p}{C_v} = 1.404 \text{ (dry air)}$$

$$\cancel{dQ} - \cancel{dW} = dH + dPE + \cancel{dKE}$$

$$dH = -dPE$$

$$hg = \Delta H = C_p \Delta T$$

C_p = Specific heat of air (J/kg K)

ΔH = rise in enthalpy (J/kg)

h = depth of the shaft (m)

g = acc. due to gravity (m/s²)

$$\text{Geothermal gradient (gg)} = \frac{0.05 \text{ (W/m}^2\text{)}}{\text{thermal conductivity (W/m}^\circ\text{C)}} \text{ (}^\circ\text{C/m)}$$

Cooling Power —

$$\text{Cooling load of a refrigerator, } q_c = q + Q_p (H_i - H_a) \text{ kW}$$

Q = quantity of air flow (m³/s)

ρ = apparent air density (kg/m^3)

h_a = allowable enthalpy of air (kJ/kg)

h_i = enthalpy of in-flowing air (kJ/kg)

Q.} The following measurements were made on VCR refrigeration system —

flow rate of air = 15 kg/s

Evaporator inlet air temperature, $t_1 = 297.3 \text{ K}$

———— outlet ————— $t_2 = 282.5 \text{ K}$

Comp. motor output = 524 kW

Calculate the cooling load & COP of the refrigeration system.

$$Q = MC (T_1 - T_2) \text{ kW}$$

M = Mass flow rate of the subsystem (kg/s)

C = specific heat capacity of the subsystem
($\text{kJ kg}^{-1} \text{K}^{-1}$)

$$\text{Given, } C = 1.005 \text{ kJ kg}^{-1} \text{K}^{-1}$$

$$\begin{aligned} Q &= 15 \times 1.005 (297.3 - 282.5) \text{ kW} \\ &= 223.11 \text{ kW} \end{aligned}$$

Coefficient of performance, $\text{COP} = \frac{\text{Heat absorbed in evaporation}}{\text{Net work supplied}}$

$$= \frac{Q_E}{W_K} = \frac{h_1 - h_3}{h_2 - h_1}$$

$$\text{COP} = \frac{223.11}{524} = 0.425$$

Airflow

Q. > Determine the nature of flow in mine airway of $2.2\text{ m} \times 1.8\text{ m}$ high. If the air velocity in mine is 0.5 m/s . Then calculate the max. vel. at which the flow still remains truly laminar.

Assume air to be at atm. pressure of 101.33 kPa and a temp. of 303 K . The kinematic viscosity of air at 303 K is $0.16 \times 10^{-4}\text{ m}^2/\text{s}$.

$$\nu = 0.16 \times 10^{-4}\text{ m}^2/\text{s} \quad \text{at } 303\text{ K}$$

$$v = 0.5\text{ m/s}$$

$$\text{Area} = 2.2 \times 1.8\text{ m}^2 = 3.96\text{ m}^2$$

$$\text{Density of air} = 1.2\text{ kg/m}^3$$

$$\text{Reynolds number, } Re = \frac{v \cdot De}{\nu}$$

$$Re = \frac{0.5 \times 1.98}{0.16 \times 10^{-4}}$$

$$= 61875$$

$$\left\{ \begin{array}{l} De = \frac{4A}{P} \\ A \rightarrow \text{Area} \\ P \rightarrow \text{Perimeter} \end{array} \right.$$

Since, $Re > 4000$, the flow is turbulent.

$$\text{Total head (m)} = \frac{P}{\rho g} + \frac{v^2}{2g} + h$$

$$\text{Total pressure (Pa)} = P + \frac{\rho v^2}{2} + \rho g h$$

Real
(viscous)

Uniform

$$P_1 + \frac{\rho_1 v_1^2}{2} + \rho_1 \cdot g \cdot h_1 = P_2 + \frac{\rho_2 v_2^2}{2} + \rho_2 \cdot g \cdot h_2$$

Non-uniform

$$P_1 + \frac{\rho_1 v_1^2}{2} + \rho_1 \cdot g \cdot h_1 = P_2 + \frac{\rho_2 v_2^2}{2} + \rho_2 \cdot g \cdot h_2 + \underbrace{F_{1-2}}_{\substack{\text{Frictional} \\ \text{loss pressure}}}$$

* In case of laminar flow of real viscous fluid —

$$\Delta P = \frac{32 \mu L v}{D^2} \quad ; \quad \begin{aligned} \mu &= \text{Dynamic viscosity (Pa/s)} \\ L &= \text{length of duct (m)} \\ v &= \text{Avg. velocity (m/s)} \\ D &= \text{Diameter of duct (m)} \end{aligned}$$

* In turbulent flow —

$$\Delta P = \frac{4 f \rho L v^2}{2D} \quad ; \quad \begin{aligned} f &= \text{friction factor} \\ \rho &= \text{density of air (kg/m}^3\text{)} \\ L &= \text{length of airway (m)} \\ v &= \text{Avg. velocity (m/s)} \\ D &= \text{diameter of airway (m)} \end{aligned}$$

* Darcy's friction factor & Reynolds number —

$$\Delta P = \frac{32 \mu L v}{D^2} = \frac{4 f \rho L v^2}{2D}$$

$$\Rightarrow f = \frac{16 \mu}{\rho v D} = \frac{16}{\left(\frac{\rho v D}{\mu} \right)} = \frac{16}{Re}$$

$$\Rightarrow \boxed{f = 16/Re}$$

$$\text{Relative roughness of airway} = \frac{\text{Mean height of surface irregularities (E)}}{\text{Diameter of opening}}$$

* Resistance Coefficient / Friction factor (f) : —

For laminar flow —

$$f = \frac{64}{Re}$$

Hagen - Poiseuille's Eqⁿ —

$$\Delta P = \frac{32 \mu L v}{D^2} = \frac{64 \mu}{\sqrt{D \rho}} \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

$$\Delta P = \frac{64}{\left(\frac{\rho \cdot v \cdot D}{\mu}\right)} \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

$$\Delta P = \frac{64}{Re} \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2}$$

→ friction factor (f)

For turbulent flow —

Blasius eqⁿ —

$$f = \frac{0.316}{Re^{0.25}} ; \text{ for } Re \leq 100,000$$

Karman - Prandtl eqⁿ —

$$\frac{1}{\sqrt{f}} = 2 \log (Re \sqrt{f} - 0.8) ; \text{ for } Re > 100,000$$

Total Pressure loss, $\Delta P = \overset{\text{(major)}}{\Delta P_f} + \Delta P_s$

↑ Friction loss due to surface drag

← shock loss due to form drag

$$\text{Friction loss } (\Delta P_f) = \frac{K P L v^2}{A} \text{ (Pa)}$$

K = friction factor (Ns^2/m^4)
 P = Perimeter of airway (m)
 L = length of airway (m)
 V = avg. velocity of airflow (m/s)
 A = cross-section of airway (m^2)

$$Q = A \cdot V$$

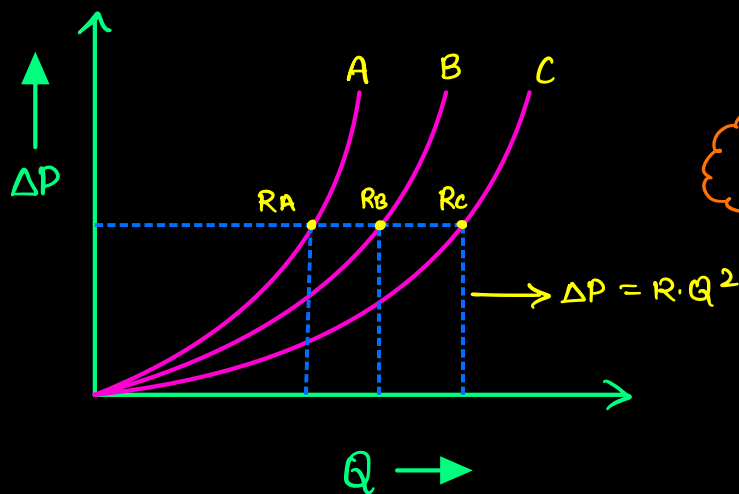
$$\Rightarrow V = \frac{Q}{A}$$

$$\Delta P_f = \frac{KPL \left(\frac{Q}{A}\right)^2}{A} = \frac{KPLQ^2}{A^2} = \frac{KSQ^2}{A^3}$$

Resistance (R)

S = Area of rubbing surface

$$\Rightarrow \boxed{\Delta P_f = R \cdot Q^2} \rightarrow \text{Square law}$$



$$R_A > R_B > R_C$$

$$\boxed{\Delta P_f = \frac{KPLV^2}{A} \cdot \frac{\rho}{1.2}} \quad ; \quad \rho = \text{density of air}$$

* Relation b/w K & f —

$$\Delta P_f = \frac{KPLV^2}{A} \cdot \frac{\rho}{1.2} = \frac{4f\rho LV^2}{2D} \quad \left\{ \begin{array}{l} D_e = \frac{4A}{P} \end{array} \right.$$

$$\frac{KPLV^2}{A} \times \frac{\rho}{1.2} = \frac{4f\rho LV^2}{2} \times \frac{\rho}{4A}$$

$$\Rightarrow \frac{K}{1.2} = \frac{f}{2}$$

$$\Rightarrow \boxed{K = 0.6f}$$

$$\Delta P_s = \Delta P - \Delta P_f$$

$$\Delta P_s \propto \Delta P_v \quad \left\{ \quad P_v = \frac{1}{2} \rho v^2 \right\}$$

$$\Delta P_s = \alpha \cdot P_v \quad ; \quad \alpha = \text{shock factor}$$

$$\Rightarrow \Delta P_s = \alpha \cdot \frac{1}{2} \rho v^2 \quad ; \quad \rho = 1.2 \text{ kg/m}^3$$

$$\Rightarrow \Delta P_s = 0.6 \alpha v^2$$

$$= \frac{0.6 \alpha Q^2}{A^2} = \frac{0.6 \alpha Q^2}{A^2}$$

$$\Delta P = \overset{R_f}{\uparrow} \frac{K_s Q^2}{A^3} + \overset{R_s}{\uparrow} \frac{0.6 \alpha Q^2}{A^2}$$

$$\Delta P = R_f \cdot Q^2 + R_s \cdot Q^2$$

$$\Delta P = (R_f + R_s) Q^2$$

$$R Q^2 = (R_f + R_s) Q^2$$

; $R_f \rightarrow \text{frictional resistance (Ns}^2/\text{m}^8)$

$R_s \rightarrow \text{shock resistance (Ns}^2/\text{m}^8)$

$$\boxed{R = R_f + R_s}$$



$$\Delta P_s = (P_1 - P_2) = \rho g (h_1 - h_2) + \frac{1}{2} \rho (v_1^2 - v_2^2) - \Delta P_f$$

Shock factor bends ———

$$\text{shock factor for normal bend, } \kappa = \frac{0.25}{R^2 \sqrt{a}} \left(\frac{2i}{\pi} \right)^2$$

$i \rightarrow$ angle of reflection

$$R \rightarrow \text{Radius ratio} = \frac{r}{W}$$

$$a \rightarrow \text{aspect ratio} = \frac{H}{W}$$

$$\text{Shock factor for right angled bend, } \kappa = \frac{0.25}{R^2 \sqrt{a}} \quad \left\{ i = \frac{\pi}{2} \right\}$$

Shock factor for a right-angled, square or circular opening —

$$\kappa = \frac{0.25}{R^2} \quad \left\{ \begin{array}{l} i = \pi/2 \\ a = H/W = 1 \end{array} \right\}$$

Shock factor for a square bend with rectangular opening

$$\kappa = \frac{0.6}{R \sqrt{a}} \left(\frac{2i}{\pi} \right)^2$$

$$\text{For right-angled bend ; } \kappa = \frac{0.6}{R \sqrt{a}}$$

$$\text{In case of square opening, } \kappa = \frac{0.6}{R}$$

Q.} In mine airway of 2.1 m wide, 2.4 m high and 600 m long contains a normal bend with a deflection of $\pi/3$ radian. Radius of curvature, $r = 3$ m. The airway passes a quantity of $9.5 \text{ m}^3/\text{s}$ of air. Calculate the pressure loss in the airway as well as resistance of the airway. Also calculate the equivalent length of a straight airway of similar size &

surface which will cause the same resistance. Assume the value of $k = 0.0098 \text{ N s}^2/\text{m}^4$.

$$W = 2.1 \text{ m}$$

$$H = 2.4 \text{ m}$$

$$L = 600 \text{ m}$$

$$Q = 9.5 \text{ m}^3/\text{s}$$

$$k = 0.0098 \text{ N s}^2/\text{m}^4$$

$$i = \pi/3 \text{ radian}$$

$$S = PL = 2(2.4 + 2.1) \times 600 = 5400 \text{ m}^2$$

$$A = 2.1 \times 2.4 = 5.04 \text{ m}^2$$

$$\kappa = \frac{0.25}{R^2 \sqrt{a}} \left(\frac{2i}{\pi} \right)^2 = \frac{0.25}{(1.42)^2 \sqrt{1.14}} \left(\frac{2 \times \pi}{3\pi} \right)^2 \quad \left\{ \begin{array}{l} R = r/w \\ a = H/w \end{array} \right\}$$

$$= 0.05$$

$$\Delta P = \Delta P_f + \Delta P_s$$

$$\Delta P = \frac{kS}{A^3} + \frac{0.6 \kappa Q^2}{A^2}$$

$$\Delta P = \frac{0.0098 \times 5400}{(5.04)^3} + \frac{0.6 \times 0.05 \times (9.5)^2}{(5.04)^2}$$

$$\Delta P = 37.4 \text{ Pa}$$

$$\Delta P = R \cdot Q^2$$

$$37.4 = R \cdot (9.5)^2$$

$$R = \frac{37.4}{(9.5)^2} = 0.414 \text{ N s}^2/\text{m}^8$$

$$\Delta P = \frac{kPL}{A^3} \cdot Q^2 \Rightarrow L = \frac{\Delta P \cdot A^3}{k \cdot P \cdot Q^2} = \frac{37.4 \times (5.04)^3}{0.0098 \times 9 \times (9.5)^2} = 601.5 \text{ m}$$

Resistance Combination ———

1.} Series Combination ———

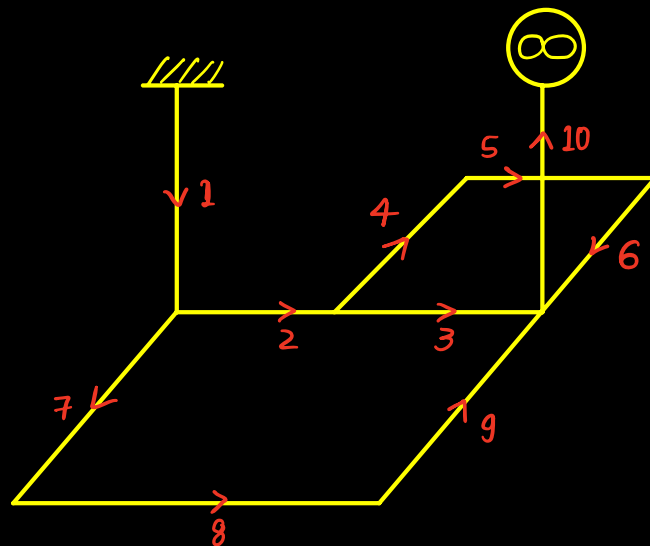
$$R_{eq.} = \sum_{i=1}^n R_i$$

2.} Parallel Combination ———

$$R_{eq.} = \left(\frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \right)^2$$

$$R_{eq.} = \left(\frac{1}{\sum_{i=1}^n \frac{1}{R_i}} \right)^2$$

Q.}



For the ventilation circuit shown in fig. following value of resistance for the individual airway are given ———

$$R_1 = 0.0559$$

$$R_5 = 0.1399$$

$$R_9 = 0.1509$$

$$R_2 = 0.1342$$

$$R_6 = 0.1453$$

$$R_{10} = 0.0447$$

$$R_3 = 0.1118$$

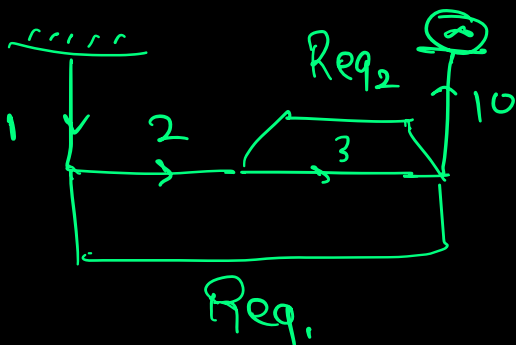
$$R_7 = 0.1062$$

$$R_4 = 0.0838$$

$$R_8 = 0.1677$$

Determine the equivalent resistance for the entire system and the total head (pressure) loss of the airway. If the fan is existing air at $47.19 \text{ m}^3/\text{s}$. Also calculate the quantity of air flowing through each airway and total air power.

A. $R_7, R_8, R_9 \rightarrow$ in series $\rightarrow R_{eq1} = 0.4248$
 $R_4, R_5, R_6 \rightarrow$ in series $\rightarrow R_{eq2} = 0.369$



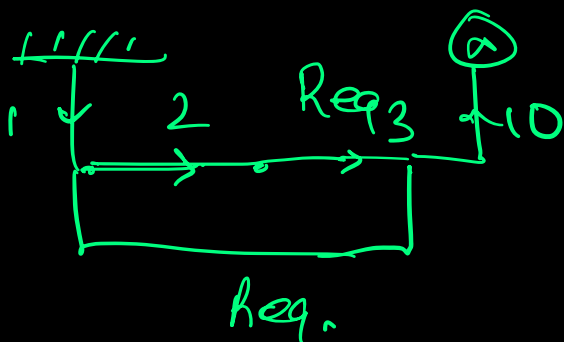
$$\frac{1}{\sqrt{R_{eq3}}} = \frac{1}{\sqrt{R_{eq2}}} + \frac{1}{\sqrt{R_3}}$$

$$R_{eq3} = \frac{R_{eq2} \cdot R_3}{(\sqrt{R_{eq2}} + \sqrt{R_3})^2}$$

$$R_{eq3} = \frac{0.04125}{0.887}$$

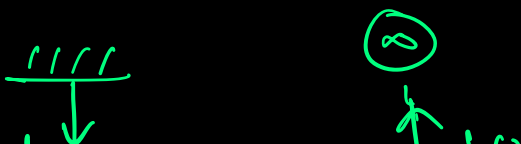
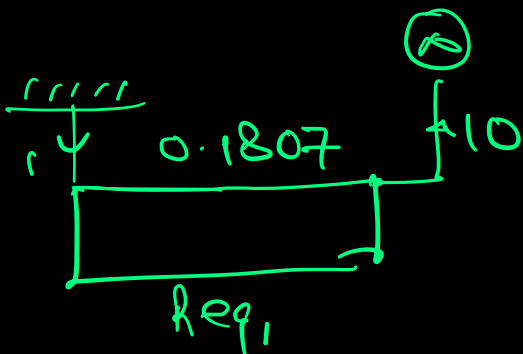
$$R_{eq3} = 0.0465$$

$$2 + R_{eq3} = 0.1807$$



$$R_{eq4} = \frac{R_{eq1} \cdot (0.1807)}{(\sqrt{R_{eq1}} + \sqrt{0.1807})^2}$$

$$R_{eq4} = 0.06619$$



$$R_{eq4}$$

$$R_{eq} = R_1 + R_{eq4} + R_{10}$$

$$R_{eq} = 0.16679 \text{ N s}^2/\text{m}^8$$

$$Q = 47.19 \text{ m}^3/\text{s}$$

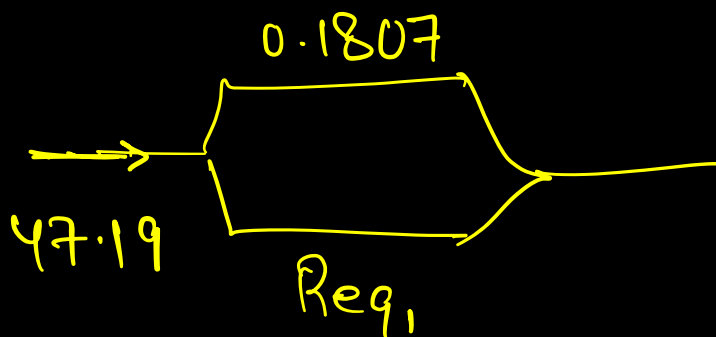
$$\text{Total Head Loss } (\Delta P) : R_{eq} Q^2$$

$$\Delta P : 0.16679 (47.19)^2$$

$$\Delta P : 371.43 \text{ Pa}$$

In Series Q + Same

$$Q = Q_1 = Q_{10} = Q_{R_{eq4}} = 47.19 \text{ m}^3/\text{s}$$



$$(Q)_{R_{eq1}} = 47.19 \sqrt{\frac{0.1807}{0.6055}}$$

$$= Q \sqrt{\frac{0.1807}{0.6055}}$$

$$(\dot{Q})_{Req_1} = 25.78 \text{ m}^3/\text{s}$$

$R_{eq_1} \rightarrow R_7, R_8, R_9$ series

$$\boxed{Q_7 = Q_8 = Q_9} = (\dot{Q})_{Req_1} = 25.78 \text{ m}^3/\text{s}$$

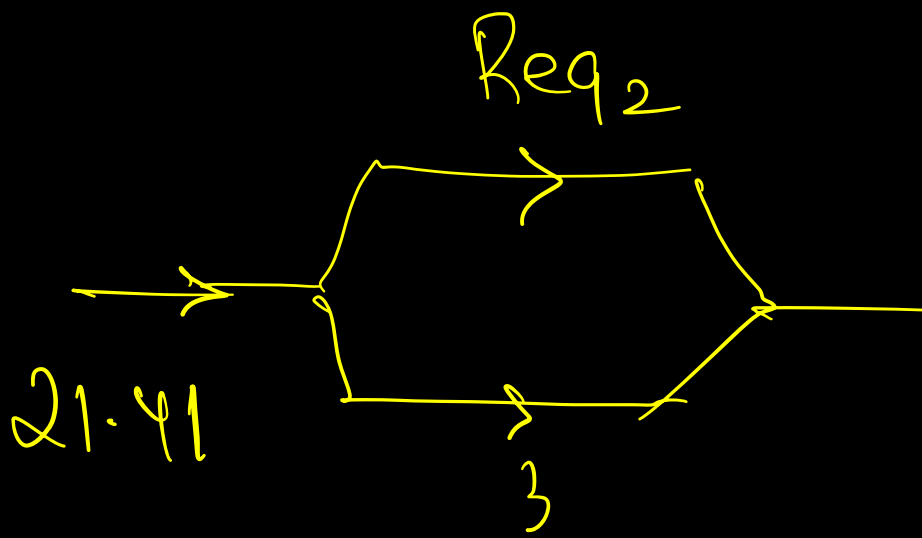
$$47.19 - 25.78 = 21.41 \text{ m}^3/\text{s}$$

↓

$$0.1807$$

$R_2, R_{eq_3} \rightarrow$ series

$$\boxed{Q_2} = (\dot{Q})_{Req_3} = 21.41 \text{ m}^3/\text{s}$$



$$Q_3 = Q \sqrt{\frac{R_{eq2}}{R_{eq2} + R_3}}$$

$$Q_3 = 21.41 \sqrt{\frac{0.369}{0.4808}}$$

$$Q_3 = 18.75 \text{ m}^3/\text{s}$$

$$\begin{aligned} Q(R_{eq2}) &= 21.41 - 18.75 \\ &= 2.66 \text{ m}^3/\text{s} \end{aligned}$$

$$Q_4 = Q_5 = Q_6 = 2.66 \text{ m}^3/\text{s}$$

$$Q_1 = 47.19 \text{ m}^3/\text{s}$$

$$Q_2 = 21.41 \text{ m}^3/\text{s}$$

$$Q_3 = 18.75 \text{ m}^3/\text{s}$$

$$Q_4 = 2.66 \text{ m}^3/\text{s}$$

$$Q_5 = 2.66 \text{ m}^3/\text{s}$$

$$Q_6 = 2.66 \text{ m}^3/\text{s}$$

$$Q_7 = 25.78 \text{ m}^3/\text{s}$$

$$Q_8 = 25.78 \text{ m}^3/\text{s}$$

$$Q_9 = 25.78 \text{ m}^3/\text{s}$$

$$Q_{10} = 47.19 \text{ m}^3/\text{s}$$

b) A horizontal underground mine gallery of 3.5 m wide, 2.5 m high and 700 m long lying at a depth of 250 m is naturally ventilated by means of two vertical shafts of 5.0 m diameter each. The gallery contains a normal bend with deflection of $\pi/3$ radians and radius of curvature of 3 m. The temperature measurements in the shafts are as follows:

Downcast shaft-top = 293 K, Downcast shaft-bottom = 296 K

Upcast shaft-top = 303 K, Upcast shaft-bottom = 304 K

The coefficient of friction k for the shafts is $0.01 \text{ N s}^2 \text{ m}^{-4}$ and for the underground gallery is $0.02 \text{ N s}^2 \text{ m}^{-4}$. If the average barometric pressure in the shafts is 101.33 kPa, calculate the velocity of airflow in the gallery due to natural ventilation. Neglect the shock loss in the shafts.

$$\text{NVP} = \frac{gDB}{287.1} \times \left(\frac{T_u - T_D}{T_u \times T_D} \right) \times 10^3 \text{ Pa}$$

B = Barometric pressure

g = 9.8 m/s^2

D = depth of the mine

T_u = Average upcast shaft temperature

T_o = Average downcast shaft temperature

Given:-

$$W = 3.5\text{m}$$

$$i = \pi/3 \text{ rad}$$

$$H = 2.5\text{m}$$

$$r = 3\text{m}$$

$$L = 700\text{m}$$

$$P = 101.33 \text{ kPa}$$

$$D = 250\text{m}$$

$$K = \begin{cases} 0.01 \text{ N s}^2/\text{m}^4 & \text{for shaft} \\ 0.02 \text{ N s}^2/\text{m}^4 & \text{for UG} \end{cases}$$

$$\text{Diameter of shaft} = 5\text{m}$$

$$\text{Average upcast shaft temperature} = \frac{303 + 304}{2}$$

$$= 303.5 \text{ K}$$

$$\text{Average downcast shaft temperature} = \frac{293 + 296}{2}$$

$$= 294.5 \text{ K}$$

$$\text{NVP} = \frac{gDB}{287.1} \times \left(\frac{T_u - T_o}{T_u \times T_o} \right) \times 100 \text{ Pa}$$

$$= \frac{9.8 \times 250 \times 101.33}{287.1} \times \left(\frac{303.5 - 294.5}{303.5 \times 294.5} \right) \times 1000$$

$$= 87.07 \text{ Pa}$$

$$\text{Cross-sectional area of level} = 3.5\text{m} \times 2.5\text{m} = 8.75\text{m}^2$$

$$\text{Perimeter of level} = 2(2.5 + 3.5) = 12\text{m}$$

$$\text{Cross-sectional area of shaft} = \frac{\pi}{4} \times 5^2 = 19.63\text{m}^2$$

$$\text{Perimeter of shaft} = \pi \times 5 = 15.70\text{m}$$

$$\Delta P = \Delta P_f + \Delta P_s$$

$$\Delta P = R_f \cdot Q^2 + R_s \cdot Q^2$$

$$\Delta P = \left(\frac{KLP}{A^3} \cdot Q^2 \right)_{\text{level}} + \left(\frac{KLP}{A^3} \cdot Q^2 \right)_{\text{shaft}} + \left(\frac{0.6 \times \alpha}{A^2} \cdot Q^2 \right)_{\text{level}}$$

$$R = \frac{\alpha}{W} = 0.86$$

$$\alpha = \frac{H}{W} = 0.71$$

$$\alpha = \frac{0.25}{R^2 \sqrt{a}} \left(\frac{21}{\pi} \right)^2$$

$$\alpha = 0.18$$

$$\Delta P = \left[\frac{0.02 \times 700 \times 12}{(8.75)^3} \right] \cdot Q^2 + \left[\frac{0.01 \times 250 \times 15.70}{(19.63)^3} \right] \cdot Q^2 + \left[\frac{0.6 \times 0.18}{(8.75)^2} \right] \cdot Q^2$$

$$87.07 = 0.25 Q^2 + 0.005 Q^2 + 0.001 Q^2$$

$$\Rightarrow 87.07 = 0.256 Q^2$$

$$\Rightarrow Q^2 = 340.117$$

$$\Rightarrow Q = 18.44 \text{ m}^3/\text{s}$$

b) A mine fan generates 500 Pa pressure and circulates 25 m³/s of air through the mine consisting of two splits, A and B. Split A and B circulates 15 m³/s and 10 m³/s of air, respectively. It is desired to increase the quantity of airflow in split B from 10 m³/s to 15 m³/s by installing a booster fan in it. Calculate the size of the booster fan if the resistance of the shafts and trunk airways is 0.2 N s²/m⁸.

Let R_A = resistance of split A

R_B = resistance of split B

R_T = resistance of shaft & trunk airway

P_B = Pressure generated by the booster fan

Q_A = Quantity of airflow in split A after installation

Q_B = Quantity of airflow in split B after installation

Q_T = Total quantity flowing through the mine after installation of booster fan

Before installation of booster fan

$$500 = R_T \times 25^2 + R_A \times 15^2 = R_T \times 25^2 + R_B \times 10^2$$

$$\Rightarrow 500 = R_A \times 15^2 + 0.2 \times 25^2$$

$$R_A = 1.67 \text{ N s}^2/\text{m}^8$$

$$\Rightarrow 500 = R_B \times 10^2 + 0.2 \times 25^2$$

$$R_B = 3.75 \text{ N s}^2/\text{m}^8$$

After installation of booster fan

$$500 = R_T Q_T^2 + R_A (Q_T - Q_B)^2$$

$$\Rightarrow 500 = 0.2 \times Q_T^2 + 1.67 (Q_T - 15)^2$$

$$\Rightarrow 500 = 0.2 \times Q_T^2 + 1.67 (Q_T^2 - 30 \times Q_T + 225)$$

$$\Rightarrow 500 = 0.2 \times Q_T^2 + 1.67 Q_T^2 - 50.1 Q_T + 375.75$$

$$\Rightarrow 500 = 1.87 Q_T^2 - 50.1 Q_T + 375.75$$

$$\Rightarrow 1.87 Q_T^2 - 50.1 Q_T - 124.25 = 0$$

$$Q_T = 29 \text{ m}^3/\text{s}$$

Now considering flow in split B ;

$$500 + P_B = R_B \cdot Q_B^2 + R_T \cdot Q_T^2$$

$$\Rightarrow 500 + P_B = 3.75 \times 15^2 + 0.2 \times 29^2$$

$$\therefore P_B = 512 \text{ Pa}$$